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Identification of Lateral Macropore Flow in a Forested Riparian Wetland through Numerical Simulation of a Subsurface Tracer Experiment

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Abstract Understanding wetland hydrogeology is important as it is coupled to internal geochemical and biotic processes that ultimately determine the fate of potential contaminant inputs. Therefore, there is a need to quantitatively understand the complex hydrogeology of wetlands. The main objective of this study was to improve understanding of saturated groundwater flow in a forested riparian wetland located on a golf course in the Lower Pee Dee River Basin in South Carolina, USA. Field observations that characterize subsurface wetland flow critical to solute transport originating from storm-generated runoff are presented. Monitoring wells were installed, and slug tests were performed to measure permeabilities of the wetland soil. A field-scale bromide tracer experiment was conducted to mimic the periodic loading of nutrients caused by storm runoff. This experiment provided spatial and temporal data on solute transport that were analyzed to determine travel times in the wetland. Furthermore, a 3-D numerical, steady-state flow model (MODFLOW) was developed to simulate subsurface flow in the wetland. A particle tracking model was subsequently used to calculate solute travel times from the wetland inlet to the outlet based on flow modeling results. It was evident that observed tracer breakthrough times were not typical of these measured wetland soil matrix conductivity values. Based on surface water sampling results at the wetland outlet, tracer arrival time was about 9 h after the injection of the tracer. These results implied an apparent mean K value of 2,050 m/day, which is 152 times larger than the mean of the measured values using slug tests (13.4 m/day). Modeling efforts clearly demonstrated this implied preferential flow behavior; particle travel times resulting from the calibrated flow model were in the order of hundreds of days, while actual travel times in the wetland were in the order of hours to a few days. This significant difference in travel times was attributed to the presence of macropores in the form of dead root channels and cavities forming a pipe-flow network. The analyses presented in this study resulted in an estimate of the ratio of matrix permeability to matrix plus macropore permeability of approximately 1/150. Eventually, the tracer test and resulting travel times between various points in the wetland were critical to understanding the true wetland flow dynamics. The final conceptual model of the hydraulic properties of the wetland soils comprised a low permeability matrix containing a web of high K macropores. Simulation of tracer transport in this system was possible using a flow model with significantly elevated K values.

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1 Introduction

In recent years, increasing attention has been focused on riparian wetlands because of their important role in regulating the fluxes of materials in ecosystems. Riparian wetlands are transitional zones between terrestrial and aquatic ecosystems in which soils and soil moisture are affected by a nearby water body. They are important natural systems for attenuating non-point source pollutant inputs from agricultural and recreational activities (Peterjohn and Correll 1984; Groffman et al. 1996; Hill 1996; Sabater et al. 2003) Hydrology of wetlands may be the most important factor in controlling wetland functions that are linked to internal geochemical and biotic processes, and involves surface- and groundwater interactions, short-circuiting, up- and downwelling of groundwater flow (Cey et al. 1999; Nwankwor and Anyaogou 2000), and flow reversals following storm events (Devito et al. 1997; Fraser et al. 2001; Elci 2003). Furthermore, the water table in most wetlands are generally not stable and fluctuate seasonally, daily or intra-daily, depending on the wetland type. The transport and fate of pollutants through wetlands is further complicated by geo- and biochemical processes. What ultimately controls the pollutant removal function of a riparian wetland is the transport velocity of the contaminants through the wetland soil and how long they are retained and exposed to removal mechanisms in the wetland system. Puckett (2004) reported that moderately deep groundwater flow paths can allow nitrate laden groundwater to pass unaltered under riparian zones and discharge into streams.

Previous studies in the literature indicate the need for a better quantitative understanding of wetland hydrology in the context of retention times and contaminant removal rates (Devito et al. 1997; Eser and Rosen 1999; Fraser et al. 2001; Maitre et al. 2003; Puckett 2004). Tracer experiments and numerical modeling can provide here useful insight into the complex nature of wetland hydrology. Reeve et al. (2000) used groundwater flow simulation results for a peatland to evaluate the importance of topography and permeability of the mineral soil underlying the peat. Hoffmann et al. (2006) conducted a detailed field and modeling study on a riparian wetland to obtain information on the relationship between groundwater residence time and nitrate removal rates. A few reports of tracer experiments dealt with hyporheic zones, wetlands and shallow aquifers (Kadlec 1994; Harvey et al. 1995; Hoag and Price 1995; Casey and Klaine 2001; Rutherford and Nguyen 2004; Parsons et al. 2004). These reports all indicate that advective transport of tracers occurs preferentially through a domain of larger pores in the sediment. These larger pores are often referred to as macropores. Macropores are known to cause non-uniform flow and preferential movement of solutes in groundwater. They can be defined as relatively large and more or less continuous voids through which rapid groundwater flow can occur.

Usually macropores in some of the cited studies above and in other similar studies in the current literature refer to vertically oriented macropores, which are located in the vadose zone and affect leaching and infiltration of solutes under variably saturated conditions. There are numerous studies investigating the influence of macropores on preferential flow of solutes in the vadose zone (e.g. Buttle and Leigh 1997; Li and Ghodrati 1997; Iqbal 1999). However, a distinction between vertical and horizontal macropores should be made, the latter being less studied in the context of contaminant transport in wetland soils under saturated conditions. Wetland soils, especially in forested wetlands, are considered to be very heterogeneous in nature, and they contain both vertical and horizontal macropores, commonly due to dead root and stem channels (Casey and Klaine 2001; Parsons et al. 2004). However, it is not well known quantitatively what the role of horizontal macropores is on solute transport through riparian wetland soil.

The present study was devoted to characterizing the hydrogeology and understanding the groundwater flow properties of a small riparian wetland (approximately 200 m long by 80 m wide) located on a golf course near Cheraw, South Carolina, USA. The effects of wetland hydrogeology on solute transport travel times were also investigated. This wetland was of interest because it receives nutrient runoff from upgradient irrigation activities. Earlier biochemical-based studies (Casey et al. 2001) for the same site indicated that active and rapidly adaptive denitrification processes were occurring in the organic wetland soils, and this remediates much of the nutrient-rich golf course runoff that flows through the wetland, and discharges into a nearby lake.

The primary objective of this study was to improve understanding of saturated groundwater flow in riparian wetland soils, which critically affects many hydrological, geochemical, and ecological functions in these ecosystems. The approach taken to meet this objective was to (1) present field observations and data that characterize subsurface wetland flow phenomena critical to solute transport originating from runoff water; (2) use a 3-D numerical model to simulate flow in the wetland and test the self-consistency of the various field observations and data; (3) track flow paths of solutes seeping into the subsurface from surface runoff and discharging back to the nearby lake located downgradient of the wetland.

2 Site Description

A natural riparian wetland was investigated that is located on the Cheraw State Park Golf Course within the Lower Pee Dee River Drainage Basin of the Sand Hills region of South Carolina, USA. This region is in the southeast of the country and in the western part of the Upper Coastal Plain of the state. There are two major hydrogeological formations underlying the site, the Middendorf and the Pinehurst formations. The Middendorf formation, which is Upper Cretaceous in age, is composed of loose to poorly indurated quartz sands, thin lenses of gritty mudstone to silty clay. The Pinehurst formation is composed mostly of dune deposits and is discontinuously dispersed across the Upper Coastal Plain (Murphy 1999). The Middendorf aquifer is the prominent aquifer at the site and is under- and overlain by clay units. It is part of the Midville aquifer system. The aquifer consists of layers of sand or high permeability limestone. It varies in extent and thickness is typically around 60 m (Aucott 1996). The primary groundwater flow direction in the aquifer is towards the southeast.

The Cheraw State Park is in Chesterfield County, and the wetland's coordinates are approximately 34° 38′13″ N latitude and 79°54′27″ W longitude. The historical-average total precipitation recorded for the site is 1,097 mm/year. The highest and lowest precipitation are typically observed in July and November, respectively. The climate for the region is mild and moderately humid with a mean temperature of 16°C. The wetland vegetation is dominated by tulip poplar (*Liriodendron tulipifera* L.), black gum (*Nyssa sylvatica* Marsh.) and red maple (*Acer rubrum* L.) along with an extensive herbaceous understory (Casey and Klaine 2001). The wetland is forested and located upgradient of Lake Juniper. It is approximately 200 m long and 30 m wide.

During storm events, the wetland receives a significant amount of surface discharge from a ditch that collects runoff with nutrients from a turfgrass putting green (730 m^2) and part of a fairway (11,000 m^2) (Fig. 1). The nutrient-rich runoff flows into an area of sandy soils directly upgradient of the wetland (wetland inlet), where most of it infiltrates the subsurface. Runoff then moves through the wetland primarily in

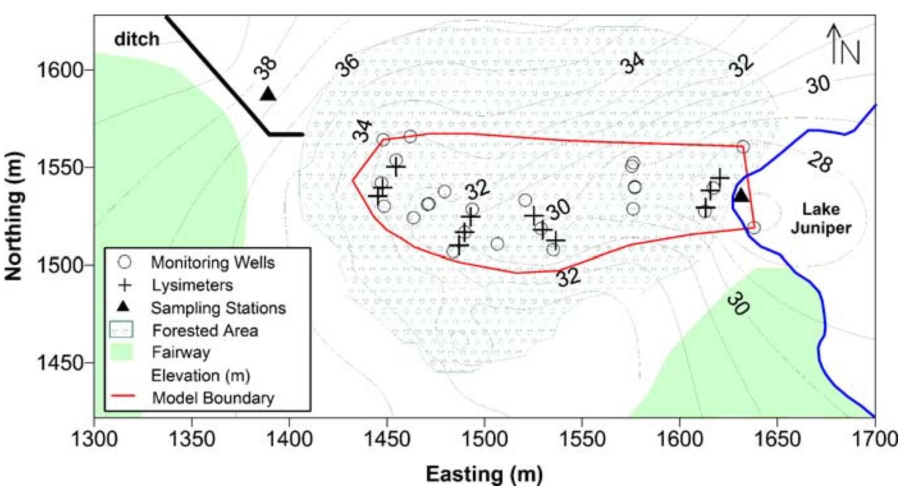

Fig. 1 Site map of studied riparian wetland with elevation contours and locations of monitoring wells, lysimeters and surface water sampling stations

the subsurface before it flows into Lake Juniper (wetland outlet). Surface water enters the wetland only during storm events. The thickness of the unsaturated zone in the wetland was relatively small; water table depth varied from zero in some central portions of the wetland to about 1.5 m at the outer boundaries. Surficial ponding and upwelling of groundwater could be observed about 140 m downgradient of the wetland inlet and 40 m upgradient of the lake.

This wetland was chosen for study because it is small in size compared to other wetlands, which made the instrumentation and monitoring less difficult and expensive. Furthermore, this wetland has already demonstrated a significant capacity for nutrient attenuation during storm events (Casey and Klaine 2001), which motivated the present study of wetland hydrogeology and solute transport.

Extensive field studies were conducted to characterize the wetland topography and stratigraphy (Elci 2003). Elevation data for the wetland land surface were obtained mainly using standard surveying techniques. Figure 1 depicts the topography of the wetland. Nineteen holes were augered within the wetland using a 6.4-cm diameter bucket auger to identify the subsurface stratigraphy. The penetration depths ranged from 0.87 to 3.2 m. Additionally, two continuous, hollow-stem auger soil borings were taken on opposite sides of the wetland to identify the deeper layers underlying the study area. Soil samples were taken using split spoons, and standard penetration tests were performed to estimate the density of the soil material. The boring on the eastern side of the wetland was sampled to a depth of 4.6 m, whereas the one on the eastern side was sampled to a depth of 10.5 m. Soil profiles were recorded for all auger holes and deep soil borings. Furthermore, in addition to the augering, a 2.4-m metal rod was pushed into the ground along several transects within the wetland to estimate the thickness of the organic top-soil layer. The penetration of the top-soil layer was determined by the difference in resistance during the pushing of the rod, and at the same time by visually confirming the presence of a denser material (sand or clay) at the bottom of the rod. Locations of auger holes and soil borings are shown in Fig. 2.

Sediments of the studied wetland could be categorized into three major soil types: organic soil (comprised primarily of peat), sandy clay and medium-coarse sand. It was observed from visual inspection of soil cores that there was no distinct impermeable layer isolating the organic layer from the deeper sediments. Below the organic soil layer, sand layers were interspersed with clay lenses (Elci 2003). The organic soil layer contained 4% to 56% volatile solids by dry weight, and it consisted mainly of a poorly drained soil with lots of plant roots, both living and dead. This layer was underlain by sandy clay and medium-coarse sand layers (Fig. 2). The thicknesses of these layers were estimated using the collected auger, boring and rod-push data. The organic soil layer thickened in the downgradient direction towards the lake. Upgradient in the wetland, a sand layer outcropped at the surface, which facilitated the infiltration of inflowing surface runoff. Another deeper sand layer was located near the upgradient edge of the wetland. In fact, the continuous soil borings indicated alternate layering of sands and sandy clays. There was also a very distinct sand layer at the downgradient edge of the wetland, as shown in Fig. 2.

3 Methods

A comprehensive characterization of the wetland hydrogeology was necessary in order to gain insight to the hydrodynamics of solute transport in wetland soils. The methods selected to meet the objectives of the study were to (1) define the wetland topography using a selected datum and land surveying techniques, (2) measure the hydraulic conductivity of the wetland soil using slug tests, (3) conduct field-scale tracer experiments to determine travel times in the wetland, and (4) set up and run a numerical groundwater flow model along with particle tracking simulations to check conceptual model and data consistency.

3.1 Construction of Monitoring Wells and Lysimeters

The monitoring network consisted of 31 4.4-cm diameter stainless steel drive-point wells and 28 lysimeter-type groundwater samplers made of PVC pipes. The drive-point monitoring wells had screens with 0.15-mm slotting. Most well screen lengths were 0.91 m, while a few well screens were 0.30 m long. The monitoring wells were manually driven into the wetland soil using a sledgehammer. They were developed by flushing water through them to loosen

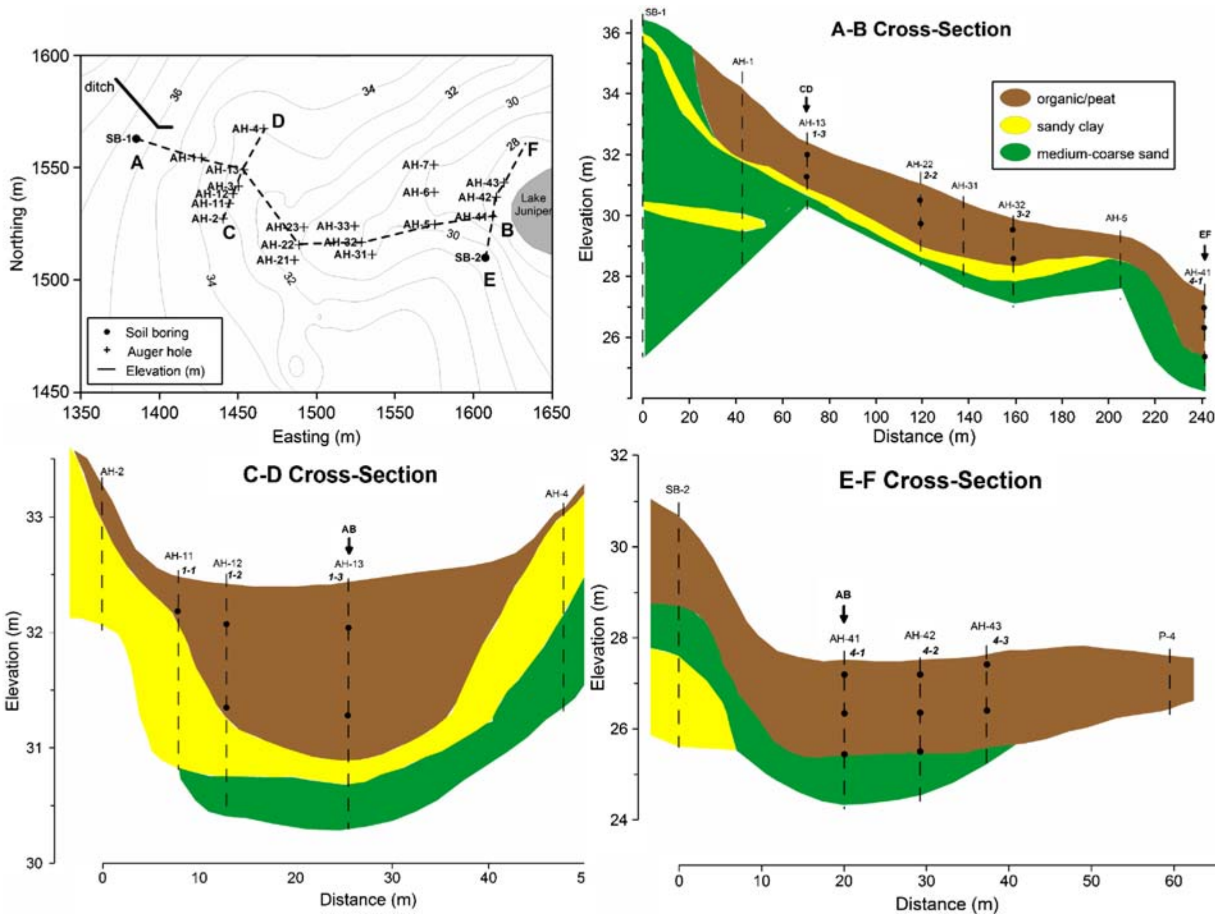

Fig. 2 Location of auger holes and stratigraphy of studied wetland: one longitudinal (*A–B*), and two transverse (*C–D* and *E–F*) cross-sections. *Italic labels* represent lysimeters names. *Filled circles* indicate elevation of lysimeter screen. Elevation axis scales are different for each cross-section

sediment that may have clogged the well screen and to ensure flow through the well screens. Twenty-five of the monitoring wells were driven to shallow to intermediate depths, ranging between 0.6 and 2.4 m, and six wells were driven deeper to depths ranging between 2.7 and 3.7 m. Well screens were below the water table at every well location. A map of the monitoring well locations is shown in Fig. 1. Most of the wells in each transect were installed in the topographic depression near the center of the wetland, where organic top-soil layer was relatively thicker. In order to determine the vertical flow field in the wetland, clustered wells with screens at different depths were installed at several locations. Installed monitoring wells were used for groundwater level measurements, slug tests, and groundwater sampling.

The lysimeters were constructed using 30 cm of 5-cm diameter PVC pipes with 0.25-mm slotting. A 5-cm PVC cap closed the bottom, and a rubber stopper with two 1-cm holes for tubing sealed the bottom of the lysimeters. Two pieces of vinyl tubing were inserted into the rubber stopper, one to withdraw the water sample, and the other for pressure relief to prevent the formation of a vacuum inside the sampler. Samplers were installed with a hand auger. The entire length of each lysimeter's screen was installed below the water table. A 2.5-cm sand pack was installed around the sampler to prevent clogging and to increase the effective radius from which the lysimeters were able to draw groundwater. Four transects of lysimeters were installed within the wetland. Three sites were located along each transect and two or three lysimeters were installed at different depths at each site. All nest sites had lysimeters at 0.30 to 0.60 m and 1.2 to 1.5 m below the ground surface. Several sites at the downgradient transects also had lysimeters at 2.1 to 2.4 m below ground surface. Lysimeters were only used to draw groundwater samples from the wetland's subsurface.

3.2 Hydraulic Conductivity Measurements

The hydraulic conductivity of the wetland soils were measured using slug tests. Slug tests were performed on 23 monitoring wells to obtain a rough estimate of the wetland soil hydraulic conductivity. Multiple tests were performed on seventeen of the wells, and test results for each well were averaged to obtain the saturated hydraulic conductivity (K_{sat}). K_{sat} values were calculated using the Bouwer and Rice (1976) method, which is used for wells in unconfined aquifers screened below the water table. The basic formulation of the method assumes isotropic and incompressible groundwater flow. As the anisotropy of the wetland soil was not known prior to the analysis, isotropic conditions were assumed and the basic formulation of the Bouwer and Rice technique was applied in the evaluation of slug test data. A theoretical study by Hyder et al. (1994) determined that anisotropy produced a considerable underestimation of K_{sat} for certain well geometries. According to their findings the ratio of the Bouwer and Rice estimate (K_{sat}) to the actual hydraulic conductivity was expected for this study to be higher than 0.45, even for an extreme anisotropy ratio of K_h/K_v=100.

3.3 Tracer Experiment

A tracer experiment was conducted in order to mimic storm-generated runoff from the surrounding turfgrass infiltrating the wetland after a storm event. In a concurrent study at the same wetland (Casey and Klaine 2001), the efficiency of the wetland for removing nutrients from storm-generated runoff was investigated. This experiment provided the opportunity to acquire detailed spatial and temporal data on solute transport, while storm runoff was occurring. The experiment was conducted by creating a stream of water from the pumping system that was used for irrigation of the golf courses. The chemistry of the runoff water was similar to the groundwater within the wetland because the irrigation system pumped its water from the nearby Lake Juniper. The stream was allowed to enter the wetland via an existing ditch that already directed turfgrass runoff from natural storm events toward the upgradient edge of the wetland. This experiment differed from natural storm events as no precipitation fell directly onto the wetland, so that all the water entered the wetland through lateral infiltration and surface flow.

The stream was amended with bromide as the tracer, which is generally conservative and is not preferentially sorbed by soils and plants (Flury and Papritz 1993). Bromide was used previously in several tracer experiment studies in wetlands (e.g. Parsons et al. 2004; Kelly and Pomes 1998). Sodium bromide was weighed and dissolved to near saturation before the amendment commenced. The concentrate was diluted up to 200 L in a carboy for delivery into the ditch by a peristaltic pump.

The tracer experiment was done under forced-gradient conditions by allowing a pre-amendment runoff to enter the wetland that lasted for 19 h. During this period, the stream water did not contain any added tracer. Following the pre-amendment period bromide was added to the inflow for 12 h. After cessation of bromide amendment, the stream of water continued to flow into the wetland to push the bromide plume further into the wetland. A constant inflow rate of 2.3 L/s was maintained during the entire tracer experiment, which lasted for 72 h.

During the daytime of the experiment, groundwater level measurements were taken manually, approximately every 2 h resulting in 22 sets of head measurements. Furthermore, lysimeters and monitoring wells were sampled at 2-h intervals during daylight on the first day of the experiment and at 4-h intervals during daylight on the following 2 days. In addition, when time permitted, surface water samples were collected within the wetland. To ensure representativeness of current groundwater conditions, monitoring wells and lysimeters were purged and samples were obtained after about 15 min. of recovery. Groundwater samples were subsequently analyzed for bromide in the laboratory. Bromide was quantified using ion-chromatography with a limit of quantitation of 0.5 mg/L.

Additional sampling stations at the inlet and outlet of the wetland were operated to obtain 30-min interval runoff samples and to continuously measure discharge. The inlet station was installed at the ditch that directs runoff to the wetland. The outlet station was set up about 20 m upstream of the lake shoreline, where groundwater upwelling to the surface occurred, thereby forming a small stream. Since the area was flat and the stream was shallow, a weir or control device to facilitate discharge measurement did not exist at the outlet, instead a stage-discharge relationship was used to measure discharge rates. Sampling at these stations provided tracer concentrations that were used to calculate tracer influx and efflux into and out of the system, respectively.

3.4 Wetland Groundwater Flow Model

3.4.1 Model Construction

A steady state, numerical, wetland flow model was developed utilizing the USGS groundwater flow code MODFLOW (Harbaugh and McDonald 1996). MODFLOW numerically evaluates the partial differential equations for groundwater flow using a block-centered finite-difference method. The wetland flow model served three purposes: (1) to improve understanding and analysis of riparian shallow surface and subsurface flow by providing flow simulations; (2) to calculate time of solute travel from the wetland inlet to the nearby lake, and (3) to estimate the apparent hydraulic conductivity of the macroporous wetland soil inferred from the tracer experiment, and compare it with simulations using field-measured hydraulic conductivities.

The model boundaries were outside the actual wetland, and they encompassed all of the saturated and seepage areas (Fig. 3). The bottom of the wetland flow model was simulated as a no-flow boundary 2.5 m below the lowest land surface with the assumption of essentially horizontal flow in the deep subsurface of the wetland. The downgradient boundary coincided roughly with the lake shoreline, and therefore was modeled as a specified head boundary. Head values for the specified head boundary were known from groundwater level measurements at a monitoring well (W21) installed right at the lake shoreline. Since there were no other hydrologic features that would coincide with a no-flow type boundary, the upgradient (west), north and south lateral boundaries were all set to general head boundaries. All assigned head values of the general head boundaries in the flow model were subject to calibration. Conductances of these boundaries were estimated based on vertically averaged hydraulic conductivity measurements (Fig. 4) and layer geometries obtained from stratigraphic data.

Evapotranspiration (ET) in the wetland was simulated using the MODFLOW Evapotranspiration package, which assumes a linear rate of water loss between a maximum ET surface and a cutoff depth. The potential ET rate for the time of the experiment was estimated as 132 mm/month using the Thornthwaite

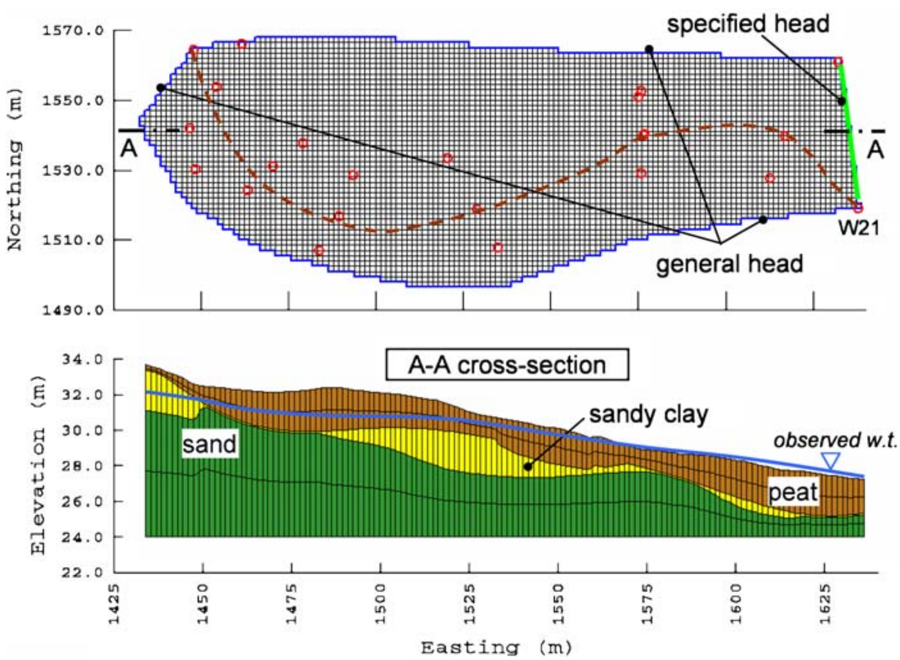

Fig. 3 Model boundaries, plan view and longitudinal cross-section of the model domain (cross-section drawn along a model row). *Circles* in plan view represent groundwater level observation points. Stream routing through the wetland is represented by the *dashed line*. W21 designates monitoring well located at the lake shoreline

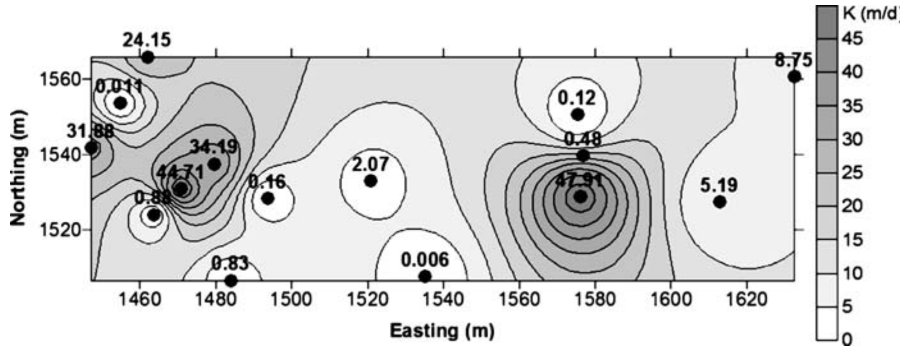

Fig. 4 Spatial distribution of K_h values calculated with the Bouwer and Rice method for shallow wells screened within 1.07 m from the ground surface

equation. This is an empirical relationship that estimates the potential ET and has been applied to wetlands with some success (Mitsch and Gosselink 2000). The estimation obtained using the Thornthwaite equation was considered as a daily averaged potential rate that warrants its use in the steady-state wetland flow model. Otherwise, if a transient flow model would be used, daily ET fluctuations needed to be taken into account.

The number of model layers was defined by the three hydrostratigraphic units: a highly organic, macroporous and very poorly drained organic/peat unit, underlain by a sandy clay and then a medium-coarse sand unit. In order to have a better vertical resolution, the organic and the sand units were further divided into two layers each. As a result, five model layers were used to vertically discretize the wetland. Layer elevations were based on kriging interpolation of stratigraphy data that was obtained during the hydrogeologic characterization phase of the study.

3.4.2 Simulation of Wetland Runoff–Groundwater Interactions

One of the difficult tasks of this modeling application was to approximate surface–groundwater interactions in a reasonable way. This task was initially approached in several ways, including the approximation of runoff as a recharge zone and the application of an additional model layer to simulate a surface water boundary, as used by Zeeb and Hemond (1998). However, the formation of ephemeral stream segments by the surface runoff entering the wetland warranted the simulation of surface-groundwater interactions with the stream-routing package of MODFLOW (Prudic 1989), and ultimately this package was selected for the model.

The runoff stream in the wetland was controlled also by geomorphologic factors, such as land surface geometry. It tended to flow through areas of convergent slopes and topographical depressions. Therefore, the stream nodes in the flow model were placed along the topographic lows of the wetland (Fig. 3). The incoming flow rate for the first segment of the runoff stream was calculated as 5,554 L/h based on the mean discharge rate measured at the monitoring station that was installed in the ditch upgradient of the wetland. Stream width was selected based on observations from the field. The Manning-type roughness coefficient was 0.112, which is typical for a rough, vegetated earth surface (Mitsch and Gosselink 2000).

3.4.3 Model Calibration

A steady state model was developed based on groundwater level measurements in the later (approximately steady state) phase of the tracer test. These groundwater levels represented a near steady-state flow condition for most of the experiment, and therefore were selected as calibration targets. The near steady-state flow condition was confirmed by verifying that the last three head measurements remained fairly constant with time. It was reasoned that such a model would serve the purposes of refining the conceptual model of the wetland and testing overall data consistency. The model was calibrated manually with a trial-and-error approach by adjusting the most sensitive model parameters until the calculated heads matched the observed heads within a calibration target of ±0.10 m, which is less than 1% of the total change in head across the modeled region, and satisfactory calibration error values were achieved. More critical model parameters were determined by a sensitivity analysis. Furthermore, the residual frequency distribution was evaluated during calibration to prevent systematic over- or underpredictions of hydraulic head.

Sensitivities of model results to changes in calibration parameters were evaluated by using the method of perturbation: a base run of the model was made and the sum of residual squares was calculated. Subsequently, calibration parameters were changed one-by-one, and the sum of residual squares re-computed. The difference in the sum of residual squares between the two runs was a measure of sensitivity. As a result, the most sensitive calibration parameters were hydraulic conductivity and specified heads along general head boundaries. The least sensitive parameters were the streambed conductances. Therefore, only hydraulic conductivities and general head boundary elevations were varied in the calibration process.

The calibration range for hydraulic conductivity was based on slug test measurements. Slug tests evaluated using the Bouwer and Rice method only measure an average K for spherical, saturated groundwater flow (K_{sat}). However, wetland sediments could be considered anisotropic due to distinctive layering. Because of uncertainty regarding anisotropy of the wetland soil, it was assumed that $K_h=K_{sat}$. The vertical K (K_v) during the calibration was calculated using arbitrarily selected anisotropy ratios.

3.4.4 Analysis of Flow Paths and Travel Times with Particle Tracking Simulations

In order to validate the calibrated flow model, particle tracking simulations were performed and particle travel times were compared to tracer arrival times from the tracer experiment. MODPATH (Pollock 1994) was used as a particle tracking model. Particles were introduced into the flow field and tracked in the continuous wetland domain according to the velocity distribution obtained from the calibrated flow model. Travel times of particles were of more interest than particle flow directions because simulated arrival times at downgradient observation points were expected to be comparable with field observations. Particles were introduced at the upgradient edge of the model domain for forward tracking, and the lake at the downgradient edge of the wetland for backward tracking. Both forward and backward particle tracking simulations were performed to obtain a range of travel times that would apply to the modeled flow field. Calculated particle travel times through the wetland were compared to measured breakthrough times of bromide from the tracer experiment. Effective porosities were selected from the literature (McWhorter and Sunada 1977) based on soil classes in the wetland, and were assigned as 0.25 for model layers 1 and 2 (peat), 0.40 for layer 3 (sandy clay) and 0.28 for layers 4 and 5 (sand).

4 Results

4.1 Measured Hydraulic Conductivity of the Wetland Soil

Hydraulic conductivity values (K_{sat}) were calculated using the Bouwer and Rice method. A contour plot of K_{sat} values for shallow screened test wells (Fig. 4) provided a hint of the highly heterogeneous structure of the wetland subsurface, where K_{sat} varied over four orders of magnitude without any spatial patterns or trends visible at the scale of measurement. The K_{sat} variation for deeper screened wells (data not shown) was similar, except the mean value of those was smaller. The range of measured K_{sat} values are given in Table 1. According to slug test result, the most conductive layer was the sand layer with a K_{sat} range of 4.0–48.0 m/day. The sandy clay layer was the least conductive layer as the range of K_{sat} was 0.005–0.16 m/day. The peat layer had a K_{sat} of 0.4–2.1 m/day.

Typical hydraulic conductivities for clean sand range from 0.8 to 110 m/day and on the other extreme for clay range from 10^{-6} to 0.1 m/day (Fetter 2001). Clearly, most of the slug test results lay within the typical range for sand, thus indicating a very permeable soil in the wetland, but without confirming that the wells were actually screened in sand. Since slug tests measured the hydraulic conductivity of the soils in the immediate vicinity of the well screen, slug

Table 1 Calibrated hydraulic conductivities compared to measured values

Layer #, soil type	Calibrated K (m/day)		Range of measured K values (m/day)
	K_h	K_z	K_{sat}
1, peat	9.6	1.9	0.4–2.1
2, peat	6.0	1.2	
3, sandy clay	3.6	1.6×10^{-2}	0.005–0.16
4, sand	48.0	4.8	4.0–48.0
5, sand	22.8	2.3	

test results should be considered as an averaged value of hydraulic conductivity for a relatively small volume of soil surrounding the well screen (Molz et al. 2005). Nevertheless, the measured values seemed reasonable, and it was expected that the subsequent transport simulations based on these estimates to be roughly consistent with the tracer tests.

4.2 Solute Transport Behavior

After initiating the tracer experiment, surface water entered the wetland from the ditch, and infiltrated into an area of predominantly sandy soils that was located at the upgradient edge of the wetland. Runoff water moved through the wetland primarily in the shallow subsurface, and a fraction of the runoff water remained on the ground surface. There was no continuous stream of surface flow throughout the wetland. However, near the wetland outlet, groundwater seeped up to the surface forming a stream that flowed into the adjacent Lake Juniper. The upwelling of groundwater here was due to a relatively strong upward hydraulic gradient. This observation is demonstrated in Fig. 5, where vertical hydraulic head gradients for various well cluster locations are shown. The hydraulic head gradient for a well cluster at the wetland outlet (W18–W36), 0.27 m/m, clearly indicated the relatively stronger upward hydraulic gradient close to the nearby lake. W36 was screened in a medium-coarse sand layer that encompassed the entire wetland. Based on the vertical hydraulic gradient measurement, it appeared that this layer recharges the wetland constantly with an increasing rate during storm events.

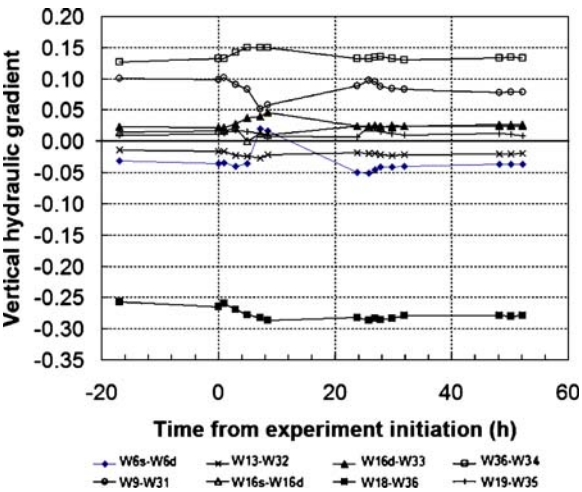

Fig. 5 Change of vertical head gradients with time during the tracer experiment for various well clusters. Negative gradients imply upward groundwater flow

Sampling stations at the inlet and outlet of the wetland were operated to obtain 30-min interval runoff samples and to continuously measure discharge. Figure 6 shows hourly-averaged discharge data for the inlet and outlet stations. The mean discharge rates during the experiment were 2.3 and 1.1 L/s for the wetland inlet and the outlet, respectively. The mean outlet discharge rate before and after the experiment was 0.39 L/s, which demonstrates the significant contribution of the upwelling to the stream at the wetland outlet. This observation of upwelling could also be supported by the measured vertical head gradient (Fig. 5). As there was no continuous stream through the wetland, discharge and vertical head gradient measurement results evinced that runoff water moved primarily through the wetland subsurface. Furthermore, sampling at these stations provided tracer concentrations that were used to calculate tracer influx and efflux into and out of the system, respectively. Based on the flux calculations, total tracer load into the wetland was 11.9 kg, of which 1.8 kg was recovered in the outlet stream during the entire sampling period. Hence it was concluded that most of the tracer remained in the wetland subsurface at the end of the sampling period. The breakthrough curves for bromide measured at the wetland inlet and outlet sampling stations are shown in Fig. 7a and b, respectively. These curves represent bromide concentrations in the stream water, noting that no continuous stream formed during the experiment. The measured breakthrough curves demonstrated that bromide moves through the wetland very rapidly. Arrival time at the wetland outlet was about 9 h after the injection of the tracer started. However, it was 11.5, 42.5, and 30 h for lysimeters 4–1, 4–2, and 4–3, respectively, which were also located close to the wetland outlet (Elci 2003). Tracer concentrations were consistently higher in the shallow subsurface at almost every location, and the tracer stream entering the wetland rapidly infiltrated the sand outcrop, therefore suggesting that bromide moved fairly rapidly between the sand and peat layers, with direction depending on the vertical hydraulic gradient. Bromide was detected at all lysimeters and monitoring wells. However, the

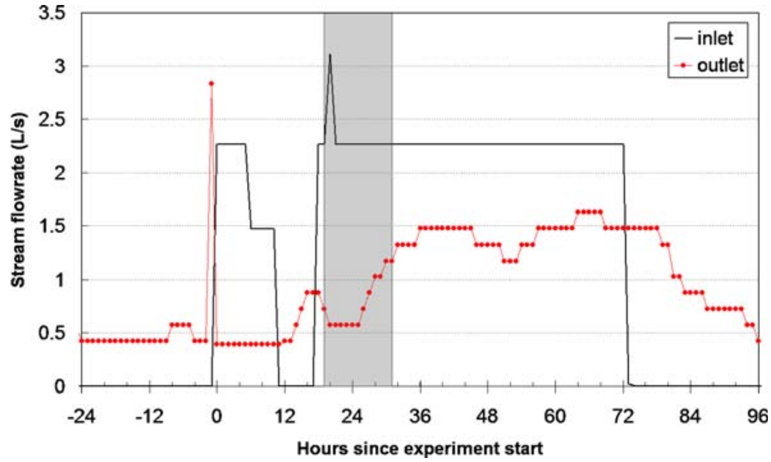

Fig. 6 Wetland stream discharge measurements. *Grey shaded area* depicts period of tracer amendment to the inflowing stream

bromide plume did not spread evenly in the lateral direction and the distribution was extremely asymmetric. Tracer concentrations were higher at the sampling sites in the northern portion of the wetland.

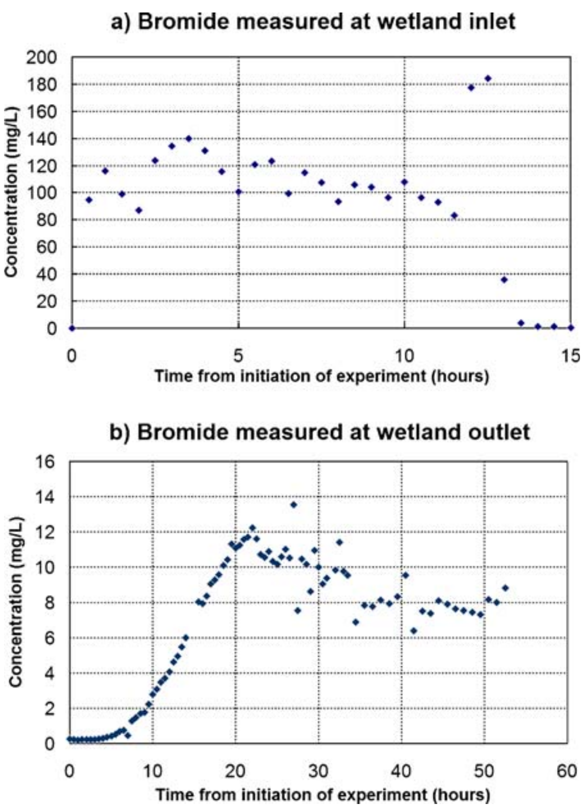

Fig. 7 Observed stream water bromide concentration **a** in the amendment stream at the wetland inlet; **b** at the wetland outlet

4.3 Flow and Particle Tracking Simulation Results

The flow model was calibrated initially based on groundwater level measurements, not tracer travel time data. After calibration, the degree of model error was satisfactory with absolute mean error and root mean squared values of 0.01 and 0.06 m, respectively. Fourteen out of 16 observation points were within the calibration target of ±0.10 m. The distribution of residuals was random and distributed about zero, and calculated water table contours matched observed water table contours very closely. In order to obtain this calibration, *K* values and specific heads along general head boundaries were varied in a reasonable range.

The most critical model parameter here was *K*, which affected simulated flow directions in the vertical, seepage velocity in the wetland, and ultimately hydraulic heads. Table 1 presents calibrated *K* values compared to measured values. Model calibrated *K* values for the first three layers were significantly higher than measured *K* values in the wetland (average factor of about 10). So the calibration that was based on water levels alone (hydraulically-calibrated) suggested that K values from the slug tests somehow did not reflect the true permeability of the wetland sediments.

Based on the hydraulically-calibrated wetland flow solution, a series of forward and backward particle tracking simulations were run. Figure 8 illustrates one of the various outcomes of particle tracking simulations, where the simulated pathlines of particles backtracked from observation points near the wetland outlet are shown. Particle pathlines in all tracking

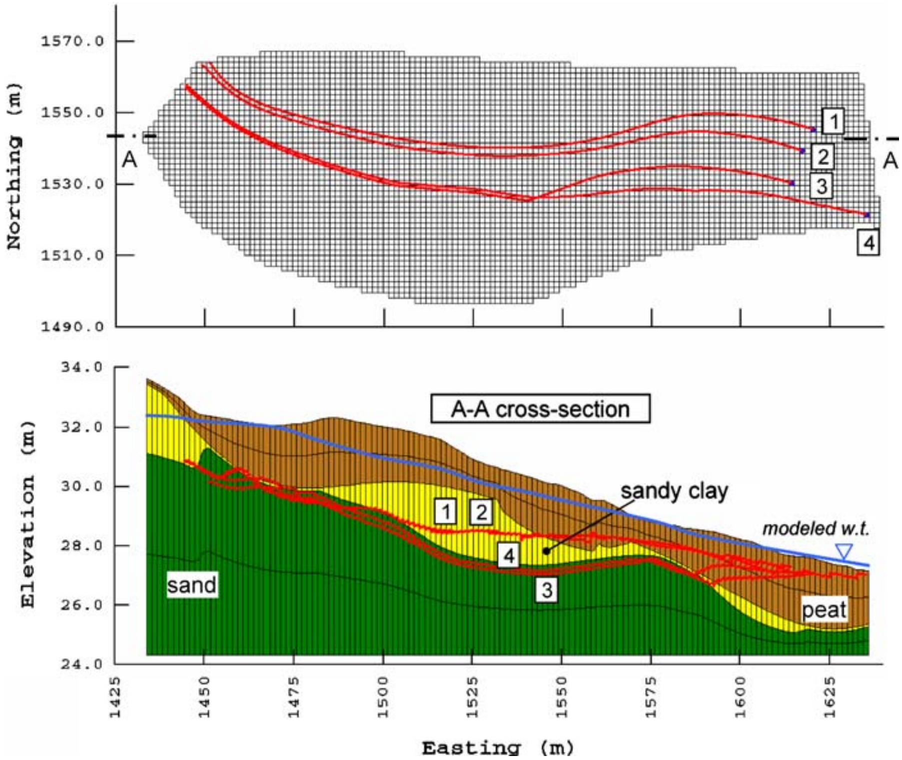

Fig. 8 Backward particle tracking simulation results—pathlines of particles released at the wetland outlet

simulations were not only seen in shallow model layers, but they intersected all model layers except the bottommost layer 5, in other words particles traveled through almost the entire vertical profile of the saturated wetland soils, while staying within 3 m below the water table. It is important to note that simulated particle pathlines were a group of solutions out of many possible pathlines that existed in reality. The range of calculated travel times for all backward-simulated particles between the wetland outlet and inlet was 70–270 days with a mean travel time of 123 days. The wide range of travel times could be attributed to particles flowing through more or less permeable layers of the model domain. Overall, this result was unexpected and was grossly inconsistent with the travel times (as low as 9 h) observed in the tracer experiments.

Despite a successful calibration of the flow model to observed water levels, particle-tracking results clearly indicated longer particle travel times, which were not consistent with tracer breakthrough times observed in the field. The bromide arrival times observed in the tracer experiment were in the order of hours to tens of hours, whereas particle travel times for the same distances, even with the larger calibrated *K* values, were in the order of tens to hundreds of days. After examining the soil more closely by visually inspecting the cores that were taken out during the augering process, we concluded that this very significant result implied enhanced conductivity of the soil due to a web of macropores and root channels in the wetland soil. In fact, the presence of macropores was also verified by Casey et al. (2004), who observed them in core samples taken for a denitrification potential assessment of the wetland soil. They pointed out that denitrifying bacterial activity localized in such macropores was more than an order of magnitude higher than in the surrounding soil matrix. Macropores with significantly elevated denitrification levels occurred throughout the wetland. The authors concluded that increased denitrification around these macropores was due to enhanced transport of nitrate, hence implying preferential groundwater flow through the horizontal macropore web in the wetland's subsurface. Others also showed that macropore-like preferential channels can rapidly move runoff and solutes through the subsurface (Waddington et al. 1993; Li and Ghodrati 1997). Based on these findings, it was plausible to believe that these macropores in the wetland were associated with an increased hydraulic conductivity of the wetland soil. Given the inconsistency of slug test results and observed tracer travel times, and also the rapid nutrient removal reported in the study reported by Casey et al. (2004), it was concluded that driving observation wells in preparation for slug tests crushed the fragile macropores, just missed them or both, thereby precluding the measurement of higher permeability values.

5 Calculation of Apparent Hydraulic Conductivities from the Tracer Test

Although the initial calibration of the flow model resulted in a close match between observed and simulated heads, it turned out after running the particle tracking simulations that the simulated average seepage velocity was very small compared to the actual velocity in the field. Therefore, it was necessary to revise and improve the flow model to obtain simulated arrival times in the wetland comparable to those observed experimentally. There were two options to improve the flow model: (1) to use smaller effective porosity, or (2) to increase hydraulic conductivity. Although both options were equally viable from a mathematical point of view, it was decided to apply the latter option since in the presence of macropores an increased overall K for the wetland would make physically more sense. Also, another study for the same wetland by Casey et al. (2004) concluded that enhanced transport of solutes through the macropores occurs resulting in elevated denitrification levels around them.

At this point, an apparent K value that represented the macroporous wetland soil, K_m, was introduced. As a first approach, K_m was determined based on a simple application of Darcy's law using averaged specific discharge values, arrival times and an average hydraulic gradient, all obtained from tracer experiment results. The specific discharge, q, for any given flow path can be defined as:

$$q = \frac{L}{t}\theta \tag{1}$$

where L is the length of the subsurface flow path from the upgradient boundary to downgradient sampling stations, t is the travel time, and θ the effective porosity. Distance (L) was estimated from the particle tracking simulation results (Table 2). Travel time (t),

Table 2 Estimates of specific discharge for particles transported through different layers of wetland soil and corresponding calculated apparent hydraulic conductivities

Pathline description	Particle no.	Pathline length, L (m)	Tracer arrival time, t (h)	Specific discharge, q (m^3/m^2 h)	Apparent hydraulic cond. K_m (m/day)
Inlet to W21	4	222	29	1.863	1,720
Inlet to 4–1–1	3	204	11.5	4.325	3,992
Inlet to 4–2–1	2	192	42.5	1.101	1,016
Inlet to 4–3–1	1	196	30	1.595	1,472

for each path line was directly obtained from the tracer data. The effective porosity of the wetland soil, θ, was estimated as an averaged value of 0.30 on the basis that simulated particles traveled through all soil types in the wetland. It must be noted that these calculations were based on a rough estimation of the wetland soil's porosity, and therefore the calculated q should be viewed as an approximate value that was further used to estimate a likely range of K_m.

Darcy's law was used to estimate the corresponding K_m values:

$$K_m = -\frac{q}{i} = -\frac{L\theta}{it} \tag{2}$$

where, i is the average horizontal hydraulic gradient observed during the tracer experiment. Based on fitting a plane to the head measurements observed during the tracer experiment, the average horizontal hydraulic gradient was calculated as $i=-0.026$. The resulting specific discharge calculated solely from field data as outlined above, and the corresponding K_m values are presented in Table 2. The arithmetic mean of all calculated K_m values was 2,050 m/day, which was 152 times larger than the mean of the measured values using slug tests (13.4 m/day). In other words, the permeability of the wetland soil including the macropores was about 150 times larger than the bulk soil permeability that was measured by slug tests.

It was evident that K_m appeared to be at least two orders of magnitude higher than the measured K from the slug tests and the calibrated groundwater flow model, which resulted in the calibrated values in the initial flow model. The calibrated K from the initial flow simulation varied from 3.6 to 48.0 m/day, as opposed to K_m values that ranged from 1,016 to 3,992 m/day.

Following this initial approximation, the wetland flow model and particle tracking simulations were re-run and validated using the apparent hydraulic conductivity, K_m. The concrete K_m values for each model layer were obtained by multiplying the calibrated K values from the initial flow simulation, which were on average higher than the field measurements obtained from slug tests, by a factor. The approach to determine a proper factor value was to match the simulated total tracer mass introduced into the system with the total bromide mass that infiltrated the wetland (ΣM=12 kg). The simulated tracer mass input was calculated by multiplying the total simulated flux at the wetland inlet boundary by the mean tracer solution concentration (93 mg/L) measured at the inlet sampling station (Elci 2003). Several trial-and-error simulations revealed that a multiplication factor of 35 gave roughly the correct amount of introduced tracer mass, and also yielded updated model K values that were within the estimated K_m range (1,016–3,992 m/day). As previously stated, particle tracking model results indicated that the tracer is likely to travel through all soil types in the wetland. Based on this finding, K values for all model layers was multiplied with the same factor to obtain K_m. Another consideration regarding the calculation of K_m is that the lack of detailed data of a possible macropore distribution required to define a variation of the multiplication factor warranted the use of a constant factor that applied for all layers, although it would be also feasible to use the model by assuming some type of distribution of macropores with depth.

Boundary conditions in the new simulation were not changed. Increasing the overall model K deteriorated the simulated head distribution slightly. The ME and RMS values for the revised model remained acceptable and were 0.03 and 0.10, respectively. Based on the updated simulation results, calculated particle travel times from the wetland outlet to the inlet were comparable to tracer travel times. Backward-simulated particle travel times resulting from the updated flow model varied between 57 and 71 h, with a mean travel time of 64 h (2.7 days).

6 Discussion

A hydrogeological characterization of a riparian wetland followed by numerical modeling of wetland subsurface flow and tracer transport was presented. Observed tracer breakthrough times were not typical of the wetland soil matrix permeability values, and they implied preferential flow behavior. Simulation efforts clearly demonstrated this type of behavior; particle travel times resulting from the calibrated flow model, based on slug tests and head data, were in the order of hundreds of days, while actual travel times in the wetland were in the order of hours to a few days. This significant difference in travel times was mainly attributed to the presence of interconnected macropores, in this case horizontally oriented root channels (mainly dead roots), cavities from decaying roots in the wetland soil, and other preferential flow channels that were crushed or sealed by monitoring well construction, or simply missed. Interconnected macropores formed some type of a pipe-flow network causing rapid transport of solutes in the wetland subsurface. Our combined measurements, slug tests and tracer tests, resulted in an estimate of the ratio of matrix permeability to matrix plus macropore permeability of approximately 1/150. Furthermore, the apparent hydraulic conductivity of the macroporous wetland soil was estimated based on field observations, providing valuable insights into riparian wetland subsurface flow. For example, such a hydraulic conductivity increase would allow for much more rapid storm water movement through the wetland and more rapid exchange between the various layers (sand, sandy clay and peat) than otherwise expected. This was consistent with the measured tracer distribution and the internal biochemical behavior and rapid stormwater renovation reported in previous studies (Casey and Klaine 2001; Casey et al. 2001).

Another finding of this study was that hydraulic conductivity measurements using slug tests during the hydrogeological characterization of the wetland were not realistic and evidently were not capable of providing the true permeability of this particular wetland soil, although the slug test may be a viable method in many other wetland types. This fact was unknown until slug tests and tracer experiments both were completed. On the other hand, higher values of K resulting from the analysis of the tracer experiment proved that rapid transport and high concentrations of the tracer were indeed captured by the monitoring wells, thereby seemingly contradicting the slug test results. Although the macropores were visible in some soil cores, it appears that most of them were either missed or crushed and sealed when wells were driven into the wetland soils. However, the tracer experiment took place about 6 months after the well installation. Therefore, it is possible that sealed macropores reconnected in this period of time to the well screens. Nevertheless, the big difference between slug test results and calibrated model hydraulic conductivities clearly indicates that the uncertainty associated with the measurements was high.

Unlike the hydraulic conductivity distribution resulting from the slug tests, the mean, steady-state, horizontal hydraulic gradient during the tracer experiment was typical of water table aquifers in sloping terrain, and was calculated as 0.026. An important point to note in passing is the potential pitfall of calibrating a model hydraulic conductivity distribution to hydraulic head data alone, unless good hydraulic conductivity data are available to start with. The problem is that a low K field and low velocity field can be consistent with the same head distribution as a high K field and a high velocity field. In the present study, the tracer test and resulting travel times between various points in the wetland were critical to understanding the true wetland dynamics.

The final conceptual model of the hydraulic properties of the wetland soils comprised a low permeability matrix containing a web of high K macropores due mainly to the presence of dead root channels. Simulation of tracer transport in this system was possible using a flow model with significantly elevated K values. The adjustment of the flow model to accommodate the early breakthrough of the tracer would be also possible using a very small effective porosity (<0.01), however such a porosity value would be too small to be realistic for a single-porosity model of the wetland. Such a system would seem to be amenable to analysis using what is called a dual porosity or dual domain transport model (Feehley et al. 2000; Flach et al. 2004). This conclusion may be also applicable to other forested riparian wetlands.

It can be concluded that the macropores were the main reason for the rapid solute transport. Processes like anion exclusion can be ruled out as they have a relatively small effect on tracer travel velocity. For example, Thomasson and Wierenga (2003) conducted a field study to investigate the spatial variability of the effective retardation factor utilizing bromide tracer experiments. The lowest retardation factor was reported as 0.45 (i.e., approximately two times faster than the bulk groundwater flow) that was indicative of anion exclusion in that case. The presence of macropores, which would not be affected by anion exclusion, would make the effect negligible as observations of this study suggest that actual seepage velocities in wetland soils with macropores are about 150 times faster than a wetland without macropores. It would not suffice to explain this difference with anion exclusion alone.

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